



## Handout 7.7: Problem Scan, Idea Filter, and Role Cards

Unit 7, Lesson 7.7. **Print pages 1 and 2 one per student. Print page 3, the role cards, one per team of three, on card stock if you have it.**

Team members: \_\_\_\_\_

Date: \_\_\_\_\_ Period: \_\_\_\_\_

## Page 1: Problem scan

Visit three of the six signs. Under each one, write problems you have **actually seen**. Not problems you imagine. Drawings count. One word counts.

### Sentence frame, four languages

| English                            | Turkish                          | Portuguese                        | Spanish                               |
|------------------------------------|----------------------------------|-----------------------------------|---------------------------------------|
| The problem is . has this problem. | Sorun . <b>Bu sorun</b> 'da var. | O problema é . tem esse problema. | El problema es . tiene este problema. |

### Sign 1: \_\_\_\_\_ (FOOD, MONEY, FAMILY, HOME, CLOTHING, or WORK)

| Problem I have seen | Who has it |
|---------------------|------------|
|                     |            |
|                     |            |
|                     |            |
|                     |            |

### Sign 2: \_\_\_\_\_

| Problem I have seen | Who has it |
|---------------------|------------|
|                     |            |
|                     |            |
|                     |            |
|                     |            |

### Sign 3: \_\_\_\_\_ (one of your three stops must be WORK)

| Problem I have seen | Who has it |
|---------------------|------------|
|                     |            |
|                     |            |
|                     |            |
|                     |            |

**Count your team's problems:** \_\_\_\_\_ (target: at least 8 per person, 12 or more per team)

## Page 2: The idea filter

### Step 1: Twenty ideas. No criticism.

Six minutes. One idea per sticky note. For each problem on page 1, what could you make or do that would solve it? **Nobody says “that’s dumb.” If you want to say it, say “later” instead.**

**Our team’s idea count:** \_\_\_\_\_ (target 20)

### Step 2: Run every idea through three questions

Cross out any idea that fails **any one** of the three. Do not argue about it. Cross it out.

|          | The question                                                                                          | Why it matters                          |
|----------|-------------------------------------------------------------------------------------------------------|-----------------------------------------|
| <b>1</b> | <b>Is the problem real?</b> Has somebody in this room actually had it?                                | If nobody has the problem, nobody pays. |
| <b>2</b> | <b>Could three middle schoolers actually build or do this in five days with what is in this room?</b> | You pitch on Friday. Not next year.     |
| <b>3</b> | <b>Would anyone pay for it, and roughly how much?</b>                                                 | Free is not a business.                 |

Write your surviving ideas here.

| Idea | Q1 real? | Q2 buildable in 5 days? | Q3 would pay? About how much? |
|------|----------|-------------------------|-------------------------------|
|      |          |                         |                               |
|      |          |                         |                               |
|      |          |                         |                               |
|      |          |                         |                               |
|      |          |                         |                               |
|      |          |                         |                               |

### Step 3: Rank the survivors

| Rank | Idea |
|------|------|
| 1    |      |

|   |  |
|---|--|
| 2 |  |
| 3 |  |

**Grade 8 line, fourth filter question:** Who is already solving this problem, and what would make somebody choose you instead?

Already solving it: \_\_\_\_\_

Why somebody would pick us: \_\_\_\_\_

**Step 4: The gate. Hand this in before you leave.**

**Our idea:** \_\_\_\_\_

**It is a product / service (circle one)**

**The FACS module it connects to** (circle at least one):

FOOD (Units 1 and 6) | MONEY (Unit 2) | FAMILY (Unit 3) | HOME (Unit 4) | CLOTHING (Unit 5) | WORK (Unit 7)

**Our one-sentence reason:**

Our idea connects to \_\_\_\_\_ because \_\_\_\_\_

**Sentence frame, four languages**

| English                               | Turkish                                | Portuguese                                 | Spanish                                     |
|---------------------------------------|----------------------------------------|--------------------------------------------|---------------------------------------------|
| Our idea connects to ___ because ___. | Fikrimiz ___ ile bağlantılı çünkü ___. | Nossa ideia se conecta com ___ porque ___. | Nuestra idea se conecta con ___ porque ___. |

**What we are physically handing a judge on Friday** (one sentence, be specific):



**Our roles:**

CEO: \_\_\_\_\_ CFO: \_\_\_\_\_ CMO: \_\_\_\_\_

**Signatures (all three):**



**Teacher initials (the gate):** \_\_\_\_\_

## Page 3: Role cards

Cut these three apart. One per team member, kept in a pocket or taped inside the team folder.

**Read this line out loud before you pick: nobody is the boss of the other two. Every role does real work, and every role's name gets said out loud on Friday.**

## CEO

Name: \_\_\_\_\_

You own the idea, the pitch, and the clock.

**Your jobs:** - Write the **business brief**: the problem, who has it, the product or service, one thing that makes it different, and the FACS module sentence. - Write **panel 5, the ask**. - Say the **hook** and the **ask** in the pitch. - **Call the time**. Twice a period, out loud: "We have \_\_\_ minutes left." - When the team is arguing instead of working, say so and pick. - On Friday: take the first judge question unless the team picked somebody else.

**On Friday a judge may ask you:** "What is your margin?" You have to know it without looking at the poster.

**You are the right CEO if:** you can keep three people moving without being mean about it, and you can make a decision when two people disagree.



## CFO

Name: \_\_\_\_\_

You own every number.

**Your jobs:** - Build the **numbers page**: unit cost line by line, price, margin per unit, fixed cost, break-even in units, one-month projection. - **Sign the numbers page.** Your signature means the arithmetic is right. - Draw **panel 4** so the numbers can be read from ten feet away. - Say the **money sentence** in the pitch. - On Friday: answer “what does one cost you and what do you charge” without hesitating.

**Your sentence to memorize:** “We stop losing money after \_\_\_ units.”

**You are the right CFO if:** you like numbers, you are careful, and you are willing to be the one who says “that does not add up.”

## CMO

Name: \_\_\_\_\_

You own the customer, the look, and the prototype.

**Your jobs:** - Run the **three customer interviews** and write the answers in the customer's **own words**, including one quote each. - Write **panels 1, 2, and 3**: the problem, the solution, the customer. - Lay out all five panels so somebody reads them in the right order. - **Build the prototype** or lead the building of it, using a skill from this year. - Say the **problem** and **who it is for** in the pitch. - On Friday: name one real person who said they would buy it, and what they said.

**You are the right CMO if:** you can talk to people you do not know, you can draw or lay something out, or you built something good this year.

*If you are stronger at drawing or building than at writing, CMO is your role and you should say so out loud.*

### All three of you

- Know **all four numbers**: unit cost, price, margin, break-even.
- **Speak part of the pitch**. How much is up to you.
- Be able to say the **FACS module sentence** from memory.

## How to pick, if nobody wants to decide

Answer these three questions out loud as a team. Do not vote. Talk.

1. **Who kept the team on time during the Unit 6 cook day, or during any group project this year?** That person is a strong CEO.
2. **Who did the arithmetic when the team had to cost something?** That person is a strong CFO.
3. **Who built, drew, sewed, or laid out the thing that came out best this year?** That person is a strong CMO.

If two people want the same role, the one who has **not** had that kind of job before takes it. You have one week left in this course and this is the last chance to do a job you have never done.